



Department of Information Technology

Academic Year 2025 – 2026(Odd Semester)

Degree, Semester & Branch: V Semester B.Tech. IT

Course Code & Title: CS3591 & Computer Networks

Name of the Faculty member (s): Mrs.M.Thulasi devi, AP/ IT

Innovative Practice Description

- **Unit / Topic:** Unit 3 / Subnetting
- **Course Outcome:** Students will gain understanding of subnetting concepts and applications.
- **Topic Learning Outcome:** TL024.
- **Activity Chosen:** Flipped Class Room
- **Justification:**
 - To assess students' comprehension level of subnetting concepts, an instructional video explaining subnet fundamentals was shared with the student cohort. This topic is frequently featured in university examinations and holds significant academic importance.
- **Time Allotted for the Activity:** 10 Minutes
- **Details of the Implementation:**
 - Learning resources were distributed to students seven days in advance via email and the learning management system (Canvas). Students were instructed to review the video content independently at home and prepare personal notes based on their comprehension.
 - During the class session, students were organized into collaborative groups. Within these teams, each participant discussed the subnetting concept along with practical examples.
 - The team comprising Mr. C. Gowtham and Mr. M. Joel Paulraj demonstrated the subnetting concept as illustrated in Figure 1.

• *CO – PO / PSO mapping:*

CO	PO1	PO2	PO3	PO4	PO5	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO3	3	3	2	2	3	1	2	2	1	2	1	1	3

(1 – Low 2 – Moderate 3 – High)

• *PO / PSO mapped:*

Innovative practice	PO1	PO2	PO3	PO4	PO5	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
	3	3	2	2	3	1	2	2	1	2	1	1	3
Justification for correlation	The students will be able to employ basic mathematical skills in applications related to the network layer.	The students could be able to formulate and analyze the various protocols for network layer mapped with real-time applications	The students could be able to design the various switching for engineering problems and real time problems	The students could be able to interpret and analyze the real-world application using Internet protocols	The students could be able to use internet addressing for data transfer	The students will complete the assessment in an ethical way	The students will work as part of a team and individually to illustrate their competencies in the network layer	The students will illustrate their abilities in communication and problem-solving as they engage with the network layer.	The students will be able to recognize the tasks necessary for developing the IPv6 protocol.	The student will identify technological developments from the past and enhance protocols within the network layer.	The students would be able to provide a solution for the address resolution protocol using their computing skills	The students could be able to provide the IT solution with network addressing with network layer	The students could be able to develop application with network layer

- **Images / Screenshot of the practice:**



Figure 1: Discussion on the topic Subnetting Arjun G

- **Reflective Critique:**

- ***Feedback of practice from students and other stakeholders:***
 - Students demonstrated active engagement throughout the activity.
 - Participants effectively showcased their understanding while explaining subnetting concepts and their application in LAN configurations.
 - This teaching method enabled students to acquire knowledge about the methodology of partitioning networks into smaller subnetworks (subnets) to facilitate direct device communication within each subnet.
- ***Benefit of the practice:*** (E.g: Learning outcome achievement improved through innovative methodology compared to traditional teaching)
 - Students developed comprehensive understanding of network subnetting principles.
 - The approach enhanced student-teacher and peer-to-peer interaction during the learning session.
 - Students experienced reduced anxiety when discussing the topic with their classmates.
 - The method promoted autonomous learning capabilities.
 - All participants exhibited thorough comprehension of subnetting concepts.
- ❖ ***Challenges faced in implementation:***
 - Several students showed reluctance to present in front of the entire class.
 - The activity exceeded the planned 10-minute timeframe, extending to 15 minutes due to initial student hesitation in approaching the presentation area.

References:

- <https://www.ritrjpm.ac.in/departments/rit-information-technology/it-innovative-practices.php>
- https://www.ritrjpm.ac.in/images/B.Tech-IT/2023-2024/IP/3_CS3352_VA_FPR.pdf
- <https://www.netacad.com/>

Signature of Faculty Member

HOD